Sam Cain

Parker, CO | (720) 717-8701 | Samca04@comcast.net
linkedin.com/in/samuel-cain-424947259 | github.com/SamCain04 | sam-cain.com

Summary

Computer Science graduate who builds privacy-focused distributed systems and applied machine learning. Experience leading small engineering teams from requirements through delivery, with hands-on work in Python, TypeScript, and AWS.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML, CSS
Frameworks and Libraries: React, Node.js, Electron, Flask, TensorFlow, pandas, NumPy, scikit-learn
Infrastructure: AWS, Docker, PostgreSQL, MongoDB, Linux, Nginx, Git, GitHub Actions, CI/CD
Security: Zero-trust architecture, post-quantum cryptography, WireGuard, OpenVPN, TLS, key custody
Practices: Agile, REST API design, data pipelines, feature engineering, model validation, technical documentation

Experience

Computer Science Capstone - StratoSplit Aug 2024 - May 2025
General Dynamics and the U.S. Coast Guard

- Led a team of 4 engineers to design and deliver StratoSplit, a multicast spatial audio system on AWS, meeting all client technical requirements on schedule.
- Implemented a zero-trust access architecture to protect sensitive communications and keep access compartmentalized.
- Wrote modular Python utilities for data handling and validation, and documented the deployment process for repeatability.

Artificial Intelligence Intern May 2024 - Aug 2024
ProAmpac

- Established the company's first AI initiative and delivered the foundational tooling that started AI adoption across IT.
- Trained 5+ employees in practical AI workflows and prompt design, improving measured departmental productivity by 30%.

Projects

EnRoute - Privacy-First VPN 2025 - Present

- Designed and built a no-logs VPN in which every tier receives identical protection and paid plans reserve capacity rather than buying stronger privacy.
- Implemented two-phase admission with client-side key custody, so private keys are generated and held on the client and never reach the control plane.
- Built the control plane in TypeScript and Node.js on PostgreSQL, plus a hardened Electron desktop client supporting WireGuard and OpenVPN transports.
- Deployed with Docker Compose behind a TLS proxy, with GitHub Actions running typechecks, tests, and container builds. Live at getenroute.net.

GPT-Hero - AI Detection Demonstration Feb 2023

- Co-developed a site exposing GPT-Zero's detection flaws across 100+ AI-generated writing samples.
- Reached over 1,000,000 unique visitors in the first month through organic TikTok distribution.

Stock Forecasting with LSTMs Jan 2024

- Built an LSTM model in TensorFlow to forecast daily stock closing prices, reaching about 70% directional accuracy in backtests.
- Engineered windowed features and walk-forward validation over real-time and historical market data feeds, tracking error against a naive baseline.

Simple Ticketing System May 2024

- Built a Flask and MongoDB web application to streamline support requests, designing the schema, REST endpoints, and authentication.
- Led a 3-person Agile team and deployed the MVP in under 4 weeks.

Education

Northern Arizona University - B.S. Computer Science Aug 2022 - May 2025
Relevant coursework: Post-Quantum Cryptography, Artificial Intelligence, Database Systems, Computer Architecture, Comparison of Learning Algorithms, Computational Theory.